L15

1 How many numbers are there smaller than 1000 that have exactly 10 factors?

5 Divide the following eight numbers into 2 groups so that the 4 numbers in each group have an equal product.

15, 19, 30, 39, 46, 57, 92, 117.

7 Divide 14, 21, 22, 26, 33, 35, 39, 55, and 65 into 3 groups so that the 3 numbers in each group have an equal product.

9 How many consecutive 0s are at the end of the product of the 100 natural numbers from 101 to 200?

1 A natural number has 16 factors. What is the minimum value of this natural number?

2 The natural numbers a and b each have exactly 5 different factors. How many different factors does ab have?

4 a has exactly 9 factors. How many factors does a^3 have at most?

10 There are 2 two-digit natural numbers whose greatest common factor is 6 and least common multiple is 90. The sum of these two numbers is _____ .

A. 96 B. 60 C. 48 D. 30

L16

7 Given that $7x \equiv 107 \pmod{11}$, the minimum value of $|x|$ is _____ .

5 Prove that $3^{2012} + 4^{2013}$ is divisible by 5.

7 There exists an integer greater than 10, such that 39, 51, and 147 have the same remainder when being divided by this integer. Find this integer.

8 What is the remainder when $1^1 + 2^2 + 3^3 + 4^4 + 5^5 + 6^6 + 7^7 + 8^8 + 9^9$ is divided by 3?

9 Given a number greater than 1, when divided by 298, 243, and 208, the remainders are a, $a + 2$, and $a + 5$, respectively. What is the value of this natural number?

L17

1 There are segments of length 2, 3, 4, 5, 6, 7, 8, and 9, and three of them are selected to form a triangle, so how many different options are there?

3 The teacher distributes 4 chocolate bars each in a different flavor to 4 of a group of 8 kids. There are _____ ways to distribute the chocolate bars.

4 How many kinds of triangles are there with lengths all integers and less than 5 on three sides? (As long as the triangles are with the same data of the length of three sides, they are the same.) _____

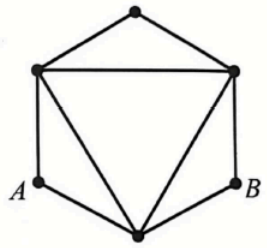
8 a, b, c are integers, and $a+b+c=12$. How many triple-digit numbers are there that meet this condition? (a, b, c can be the same)

A. 52 B. 53 C. 54 D. 55

1 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(3) If exactly one of A, B is picked, there are _____ ways of picking.

4 As shown in the picture, start from the point A, walk along the line sections, and finally reach the point B.



(1) If pass each black dot once, how many ways to go?

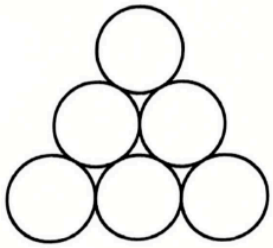
(2) If pass each black dot at least once, how many ways to go?

5 There is a batch of thin wooden strips with lengths of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11 cm, and they are numerous enough to select 3 as three sides to enclose a triangle. If one of the sides is specified to be 11 cm long, how many different triangles can you have? _____

6 Call a positive integer monotonous if it is a one–digit number or its digits, when read from left to right, form either a strictly increasing or a strictly decreasing sequence. For example, 3,23578, and 987620 are monotonous, but 88,7434, and 23557 are not. How many monotonous positive integers are there? (2017 AMC 10B Problems, Question #17)

A. 1024 B. 1524 C. 1533 D. 1536 E. 2048

7 In the figure below, 3 of the 6 disks are to be painted blue, 2 are to be painted red, and 1 is to be painted green. Two paintings that can be obtained from one another by a rotation or a reflection of the entire figure are considered the same. How many different paintings are possible? ( ) (2017 AMC 12B Problem, Question #13)



A. 6 B. 8 C. 9 D. 12 E. 15

L18

1 A bag contains 4 balls that are identical in size and texture, including 2 red balls and 2 green balls. Two balls are randomly drawn one after another without replacement. What is the probability that the two balls are of the same color? ( ) .

A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

3 If one number is chosen at random from all the positive divisors of 2025, what is the probability that this number is a perfect square?

A. 0.4 B. 0.3 C. 0.6 D. 0.7 E. 0.8

4 Randomly choose an integer from 1 to 6, the probability distribution follows:

$P(6) = P(1)$;

$P(2) = P(4)$;

$P(3) = 3P(5)$;

$P(6) = 2P(4) = 4P(5)$.

Find $P(1)$.

5 Assign volunteers A, B, C to classrooms numbered $1, 2, 3$ for cleaning, with exactly one volunteer in each classroom. What is the probability that A is not assigned to classroom 3?

A. $\frac{2}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{3}$ E. $\frac{3}{5}$

6 To assess students' physical fitness, 3 items are chosen at random from a total of 7 events: 3 track and field events, 2 ball games, and 2 martial arts events. What is the probability that the chosen 3 items come from exactly two categories?

A. $\frac{22}{35}$ B. $\frac{2}{7}$ C. $\frac{4}{7}$ D. $\frac{11}{35}$ E. $\frac{33}{35}$

7 If three vertices are chosen at random from the eight vertices of a cube and connected to form a triangle, what is the probability that the triangle is equilateral?

A. $\frac{1}{7}$ B. $\frac{1}{14}$ C. $\frac{2}{7}$ D. $\frac{4}{35}$ E. $\frac{5}{14}$

2 One fair die has faces $1, 1, 2, 2, 3, 3$ and another has faces $4, 4, 5, 5, 6, 6$. The dice are rolled and the numbers on the top faces are added. What is the probability that the sum will be odd?

(2005 AMC 10B Problem, Question #9)

A $\frac{1}{3}$ B. $\frac{4}{9}$ C. $\frac{1}{2}$ D. $\frac{5}{9}$ E. $\frac{2}{3}$

6 Suppose that one of every 500 people in a certain population has a particular disease, which displays no symptoms. A blood test is available for screening for this disease. For a person who has this disease, the test always turns out positive. For a person who does not have the disease, however, there is a 2% false positive rate - in other words, for such people, 98% of the time the test will turn out negative, but 2% of the time the test will turn out positive and will incorrectly indicate that the person has the disease. Let p be the probability that a person who is chosen at random from this population and gets a positive test result actually has the disease. Which of the following is closest to p?

(2012 AMC 10B Problem, Question #18)

A. $\frac{1}{98}$ B. $\frac{1}{9}$ C. $\frac{1}{11}$ D. $\frac{49}{99}$ E. $\frac{98}{99}$

7 A box contains three bags. One bag contains one white pebble and three black pebbles. Another bag also contains one white pebble and three black pebbles. The remaining bag contains one white pebble and four black pebbles. Eddie reaches into the box and randomly pulls out a bag, then reaches into the bag and randomly pulls out a pebble. What is the probability that this pebble is white?

A. $\frac{7}{30}$ B. $\frac{3}{13}$ C. $\frac{1}{9}$ D. $\frac{1}{4}$ E. $\frac{11}{36}$

8 Forty slips are placed into a hat, each bearing a number 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, with each number entered on four slips. Four slips are drawn from the hat at random and without replacement. Let p be the probability that all four slips bear the same number. Let q be the probability that two of the slips bear a number a and the other two bear a number $b \neq a$. What is the value of $\frac{q}{p}$?

(2005 AMC 10B Problem, Question #21)

A. 162 B. 180 C. 324 D. 360 E. 720